

Unit 2: Kotlin Fundamentals

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```
fun main() {  
  
    // Exercise 1: Movie tickets  
    val child = 5  
    val adult = 28  
    val senior = 87  
  
    val isMonday = true  
  
    println("The movie ticket price for a person aged $child is \${ticketPrice(child, isMonday)}.")  
    println("The movie ticket price for a person aged $adult is \${ticketPrice(adult, isMonday)}.")  
    println("The movie ticket price for a person aged $senior is \${ticketPrice(senior, isMonday)}.")  
  
    println()  
}
```

```
// Exercise 2: Temperature conversion  
printFinalTemperature(27.0, "Celsius", "Fahrenheit") {  
    9.0 / 5.0 * it + 32  
}  
  
printFinalTemperature(350.0, "Kelvin", "Celsius") {  
    it - 273.15  
}  
  
printFinalTemperature(10.0, "Fahrenheit", "Kelvin") {  
    (it - 32) * 5.0 / 9.0 + 273.15  
}  
  
println()
```

```
// Exercise 3: Person  
val amanda = Person("Amanda", 33, "play tennis", null)  
val atiqah = Person("Atiqah", 28, "climb", amanda)  
  
amanda.showProfile()  
atiqah.showProfile()  
  
println()
```

```

// Exercise 4: Phone
val phone = Phone()

phone.checkPhoneScreenLight()
phone.switchOn()
phone.checkPhoneScreenLight()
phone.switchOff()
phone.checkPhoneScreenLight()

println()

// Exercise 5: Auction
val winningBid = Bid(5000, "Private Collector")

println("Item A is sold at ${auctionPrice(winningBid, 2000)}.")
println("Item B is sold at ${auctionPrice(null, 3000)}.")
}

```

```

// -----
// Exercise 1
// -----

fun ticketPrice(age: Int, isMonday: Boolean): Int {
    return when {
        age <= 12 -> 15
        age >= 65 -> 20
        isMonday -> 25
        else -> 30
    }
}

// -----
// Exercise 2
// -----

fun printFinalTemperature(
    initialMeasurement: Double,
    initialUnit: String,
    finalUnit: String,
    conversionFormula: (Double) -> Double
) {
    val finalMeasurement =
        String.format("%.2f", conversionFormula(initialMeasurement))

    println(
        "$initialMeasurement degrees $initialUnit is " +
        "$finalMeasurement degrees $finalUnit."
    )
}

```

```
// -----  
// Exercise 3  
// -----  
  
class Person(  
    val name: String,  
    val age: Int,  
    val hobby: String?,  
    val referrer: Person?  
) {  
    fun showProfile() {  
        println("Name: $name")  
        println("Age: $age")  
        println("Likes to $hobby.")  
  
        if (referrer != null) {  
            println(  
                "Has a referrer named ${referrer.name}, " +  
                "who likes to ${referrer.hobby}."  
            )  
        } else {  
            println("Doesn't have a referrer.")  
        }  
    }  
}
```

```

// -----
// Exercise 4
// -----

class Phone(var isScreenLightOn: Boolean = false) {

    fun switchOn() {
        isScreenLightOn = true
    }

    fun switchOff() {
        isScreenLightOn = false
    }

    fun checkPhoneScreenLight() {
        val phoneScreenLight =
            if (isScreenLightOn) "on" else "off"

        println("The phone screen's light is $phoneScreenLight.")
    }
}

// -----
// Exercise 5
// -----

class Bid(
    val amount: Int,
    val bidder: String
)

fun auctionPrice(
    bid: Bid?,
    minimumPrice: Int
): Int {
    return bid?.amount ?: minimumPrice
}

```

Code Output:

The movie ticket price for a person aged 5 is \$15.
The movie ticket price for a person aged 28 is \$25.
The movie ticket price for a person aged 87 is \$20.

27.0 degrees Celsius is 80.60 degrees Fahrenheit.
350.0 degrees Kelvin is 76.85 degrees Celsius.
10.0 degrees Fahrenheit is 260.93 degrees Kelvin.

Name: Amanda
Age: 33
Likes to play tennis.
Doesn't have a referrer.
Name: Atiqah
Age: 28
Likes to climb.
Has a referrer named Amanda, who likes to play tennis.

The phone screen's light is off.
The phone screen's light is on.
The phone screen's light is off.

Item A is sold at 5000.
Item B is sold at 3000.